PROBLEM STATEMENT

In today’s data-driven food service industry, restaurants and online ordering platforms generate large volumes of transactional data, including order details, item names, quantities, and pricing information. The Chipotle dataset represents such real-world data, where multiple customers place orders consisting of various menu items, often resulting in repeated entries and complex data patterns. Analyzing this growing dataset efficiently poses a significant challenge using traditional data processing techniques.

The dataset contains valuable information such as order IDs, item names, quantities, and item prices. However, extracting meaningful insights—such as identifying the most popular items, calculating total revenue, or analyzing ordering patterns—is difficult due to the size and structure of the data. As the dataset grows, conventional methods become inefficient, leading to increased processing time and reduced scalability.

This project aims to overcome these challenges by implementing a scalable Big Data solution using the Hadoop ecosystem. By leveraging HDFS for distributed storage and MapReduce for parallel processing, the system efficiently processes large transactional datasets to compute key metrics such as total sales, item popularity, and order trends.

Ultimately, this project demonstrates how distributed computing can transform raw transactional data into actionable insights, enabling businesses to optimize operations, improve decision-making, and enhance customer satisfaction in a competitive environment.